Project Proposal

1. Title and Team formation

Title: Enhancing Linear Classifiers with Contextual Embeddings for Sentiment Analysis

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2. Motivation and Hypothesis

The problem we are trying to solve is that traditional text classifiers are fast but lack an understanding of deep language context. On the other hand, context-analysis models are very accurate in sentiment analysis but computationally are very slow. We propose a hybrid model which leverages the strengths of both approaches.

We hypothesise that incorporating deep contextualised word embeddings from a pretrained model into a simple linear classifier will significantly improve its accuracy on sentiment tasks, closing the gap with massive neural networks whilst remaining computationally efficient.

3. Methodology

Our project will be implemented in Python using standard machine learning and NLP libraries. The experiment will consist of four main stages to ensure a clear, repeatable experiment:

(i) Dataset selection and Preprocessing: We will be using a focused, publicly available dataset for binary sentiment analysis (eg. IMDb Movie Review Dataset or a target social media corpus). The text data will be cleaned and split into standard training, validation, and test sets.

(ii) Baseline Implementation: As a control, we will be implementing a standard linear classifier, specifically Logistic regression and a Perceptron. These baseline models will be trained using a standard sparse feature representation method (such as TF-IDF word counts).

(iii) Contextual Feature Extraction: To incorporate the extra source of knowledge into the feature design of our model, we will be using a pretrained transformer model. Keeping the transformer's weights frozen to minimise computational overhead, we'll perform a forward pass with our training data and validation data to extract dense contextual embeddings from the model's hidden states.

(iv) Hybrid Model Training and Validation: We will be replacing the traditional TF-IDF inputs for our newly extracted embedding matrices. The simple linear models will be trained on these dense representations. During validation, the unseen validation text will be passed through the same frozen transformer pipeline to generate embeddings before being evaluated by the classifiers.

4. Dataset

As to comply with the project's rules, our data will be consisting only of text; we have decided to use the IMDb dataset as it contains a bit more than 50,000 datapoints. To have a more representative conclusion, we will attempt to use the whole dataset in this project, using an 80-10-10 split for training, validation, test respectively. As the computation of the embeddings for such a large set would be hugely costly in terms of RAM, we will implement a batching technique to accommodate for it.

5. Evaluation Plan

All models will be evaluated on the following metrics: (i) accuracy, (ii) training time, and (iii) computational complexity. We will compare the baseline model against the hybrid model to quantify the impact of the contextual embeddings. Furthermore, if times allow it, we will also be comparing those with the unfrozen pretrained language model as to see how much accuracy we'd be losing with our hybrid model.